

Cool Tweak

Install Ubuntu 10.04 From A USB Drive Without BIOS Support for USB Boot

Tired of burning CDs each time you want to try out the latest Ubuntu release because your not-so-new computer doesn't support booting from the USB? Don't worry, now you can use your USB drive as a boot medium, and install newer versions of Ubuntu even if your BIOS doesn't support a USB boot!



For my experiment, I used an Acer Travelmate 290L series laptop made in 2005, with no BIOS support for a USB boot, and a Kingston Traveller 1 GB pen drive. The laptop had Ubuntu 9.04 installed on it, and I used this to create the USB start-up disk from a Ubuntu 10.04 ISO, which I downloaded from <http://www.ubuntu.com/desktop/get-ubuntu/download>.

Here is the procedure I followed:

1. Download the Ubuntu 10.04 ISO.
2. Back up the contents of the USB pen drive, and delete all the files/folders on it.
3. Use the USB Startup Disk Creator utility (accessible at *System > Administration > Startup Disk Creator*) to create a USB boot disk on the pen drive from the Ubuntu 10.04 ISO. Figure 1 shows how the Startup Disk Creator utility appeared on my system.
4. When the Startup Disk Creator utility is done completing the boot disk, create a *usb-boot* sub-directory under */boot* on your running Linux system:

```
sudo mkdir -p /boot/usb-boot
```

5. Copy the *vmlinuz* & *initrd.lz* files from the USB drive to the newly created */boot/usb-boot* folder. Figure 2 shows what you

may expect to see if you open a file browser window to the *casper* sub-directory of the mounted USB drive.

```
sudo cp /media/KINGSTON/casper/* /boot/usb-boot/
```

Note: *KINGSTON* is the label (and thus, the automatic mount-point under */media*) of the filesystem on the USB drive I used. Depending on the brand of the USB drive you're using (or the partitioning/format options it was subjected to), the label may differ—and hence the mount point, and so the path above may need to be changed.

6. Add the following boot stanza to */boot/grub/menu.lst* (use *sudo gedit /boot/grub/menu.lst* to edit it as root):

```
title Run Ubuntu 10.04 from USB FLASH DRIVE
root (hd0,4)
kernel /boot/usb-boot/vmlinuz file=/cdrom/preseed/ubuntu.seed
boot=casper noprompt cdrom-detect/try-usb=true
persistent quiet splash $!$$.modest=1
initrd /boot/usb-boot/initrd.lz
boot
```

Note:

- (*hd0,4*) is the location where the current OS (Ubuntu 9.04) is installed.